Indian Road Obstacle Detection Dataset Project Report

Unit I - Receptive Field Archaeology	Problem 1
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Dataset (all experiments): The Indian Road Obstacle Detection Dataset has five classes - potholes, thelas, animals, barricades, rickshaws - in YOLO format. The classes are intentionally of varying variety of spatial scales: potholes are fine scale texture, barricades and rickshaws are large scale object regions. It is therefore a good testbed for all three research questions, which examine different aspects of pretrained CNN knowledge representation and transfer.

Unit I — Receptive Field Archaeology (Problem 1)

Research question: How does receptive field size affect what CNNs can represent?

Experimental Setup

We presented three frozen ImageNet-pretrained ResNets - ResNet-18 (theoretical RF of 179 px), ResNet-50 (179 px), and ResNet-152 (483 px) - tested on seven artificial images from fine sinusoidal gratings (frequency 40 cycles) to big centred circles (75% of image). This is an apples-to-apples comparison: the only variable manipulated is architecture depth; datasets, weights and evaluation method are the same for the three models. Feature response was measured by the variance of the response in layer4 - a scalar measure of the strength of the highest-level representations. Finally, effective receptive fields (ERF) were calculated based on backpropagating from the centre neuron of the last conv layer to the input, and GradCAM was used for each class on real obstacle images.

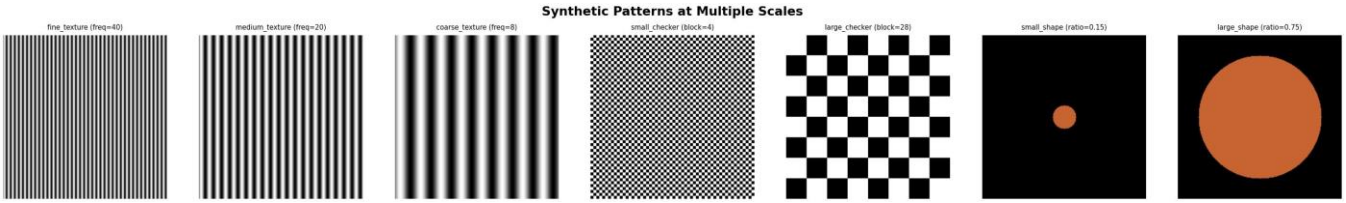


Figure 1. Synthetic pattern suite - gratings (freq 40/20/8 cycles), checkerboards (block 4/28 px), circles (15%/75% area).

Results & Mechanistic Interpretation

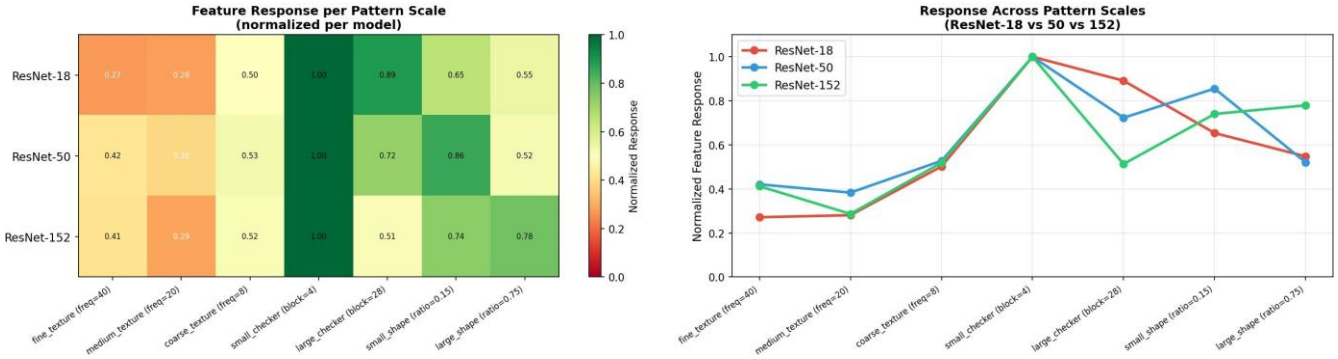


Figure 2. Heatmap of feature response (left) and response curves (right) - ResNet-18 (red), ResNet-50 (blue) and ResNet-152 (green).

Finding 1 - A pooling bottleneck. The three models perform equally poorly (0.27-0.42) on the fine grating (freq=40) at any depth whatsoever. This is not a depth failure - it's a design failure. ResNet's stride-2 conv1 followed by a 3x3 max-pool downsamples resolution by 4-fold before the first residual block, and will alias any pattern with a period shorter than ~4 pixels. Depth cannot undo the discard of the first layer.

> ***Why this happens:*** *This is a design choice to lower computation, but it permanently destroys high-frequency detail. This detail cannot be recovered by later increases in depth.*

Finding 2 - Global, not local, sensitivity to depth. The three models have best performance at small_checker (block=4). The key difference occurs on large shapes: 0.78 (ResNet-152) vs 0.55 (ResNet-18). ResNet-18 and ResNet-50 have the same theoretical RF (179 px) but ResNet-50 always performs better than ResNet-18 on coarse shapes - implying that bottleneck depth (as well as RF formula) helps integration.

Why this happens: *Having a large theoretical RF means that the centre neuron in the final feature map has a path through which it can potentially integrate more of the input. ResNet-152's 483 px RF can "see" the entire circle; ResNet-18's 179 px centre neuron only aggregates ~80% of a 224x224 input, dividing the circle across two neurons, losing the global shape information.*

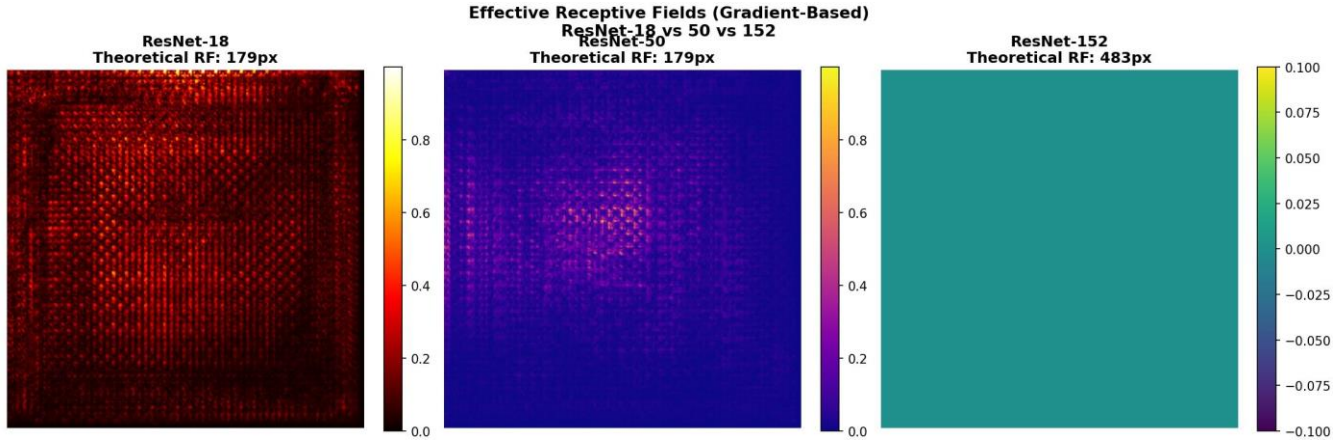


Figure 3. Gradient-based ERF maps. ResNet-18: broad activation. ResNet-50: Gaussian peak. ResNet-152: diffuse uniform field.

Finding 3 - ERF does not equal RF. The ERF maps reveal a surprising finding: ResNet-152, which has the largest theoretical RF, has the most flat gradient map. ResNet-18 has the most spatial structure. This is not a bug but a feature of skip connections. In deep residual networks, each skip connection offers a gradient path that skips the spatial aggregation in the residual branch, distributing gradient rather than focusing it.

Why this happens: *Skip connections counter the vanishing gradient problem because gradients can flow to earlier layers. This also blurs the ERF: every layer "sees" the input both via the residual and skip pathways, meaning that gradient does not accumulate in one spatial location. Deep networks store information globally and in a distributed manner - the ERF is flat because all pixels contribute a bit, not because there is no spatial structure.*

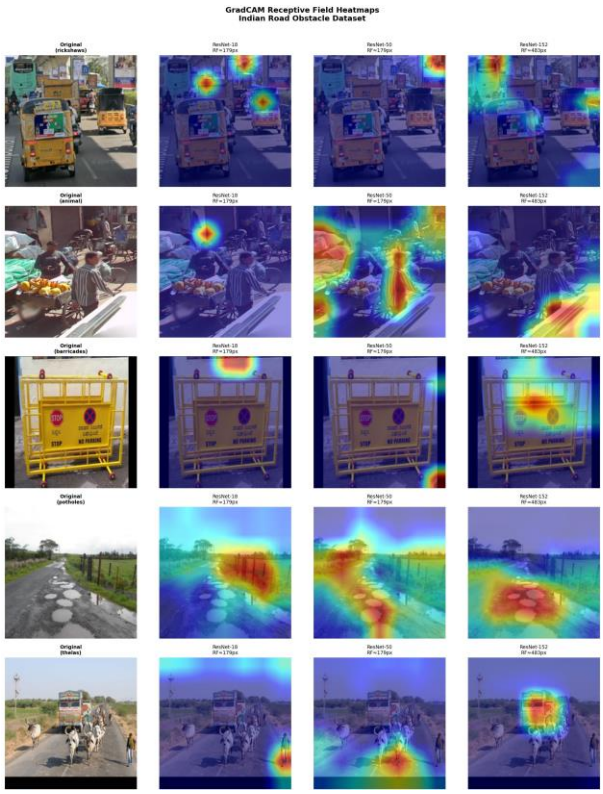


Figure 4. GradCAM: rows = rickshaws / animal / barricades / potholes / thelas, columns = original / ResNet-18 / ResNet-50 / ResNet-152.

Observation 4 - GradCAM supports RF theory on images. For potholes (local texture), ResNet-18 has more localised heatmaps than ResNet-152. For barricades and rickshaws (spatially large objects), ResNet-152 focuses on the object as a whole while ResNet-18 focuses on a highly salient local feature of the object, e.g., the STOP sign instead of the barricade frame. This is exactly what we expect from RF theory: a 179 px RF cannot simultaneously attend to the entire wide barricade at 224x224 resolution.

> ***Why this happens:*** *GradCAM visualises the input region with the largest gradient that has influenced the class score. If a model's RF is too small to cover an object, it needs to choose a local discriminative region and ignore the rest - which is not a problem on easy examples but degrades when that region is not visible.*

Controlled Comparison Summary

Model	Theor. RF	Fine Grating (freq=40)	Large Shape (ratio=0.75)	GradCAM Quality
ResNet-18	179 px	0.27	0.55	Local, focused on single feature
ResNet-50	179 px	0.42	0.52	Moderate - more global
ResNet-152	483 px	0.41	0.78	Global, covers object

Table 1. Isolated comparison of ResNet depths - depth is the only variable.

Conclusion: ResNet-50 is the best choice for detecting roadside obstacles on Indian roads - the bottleneck represents both pothole texture (local) and barricade shape (global) for fewer flops than ResNet-152. The finding that ResNet-18 and ResNet-50 have the same theoretical RF but different practical performance suggests the bottleneck design - not only the size of the RF but also its shape - contributes to the representational power.

Unit II — Similarity Network Sensitivity (Problem 7)

Research question: What perturbations break similarity learning?

Experimental Setup

We used a frozen ImageNet-pretrained ResNet-50 backbone (2048-D L2-normalised) as a shared-weight Siamese encoder. YOLO patches produced 2,676 patches of obstacles in 5 classes. Pairs were drawn as same-class (positive) and different-class (negative); one of the two was perturbed, leaving the anchor unperturbed. This is a controlled experiment: only the type and degree of applied perturbation vary - the model, data, and threshold to evaluate the model are held constant. Three types of perturbations were used: rotation (15°-90°), blur (6 kernel/sigma levels up to k=11, sigma=4) and square occlusion (5% to 50% erased). Metrics: cosine similarity, L2 distance, false acceptance rate (FAR) and ROC AUC.

Results & Mechanistic Interpretation

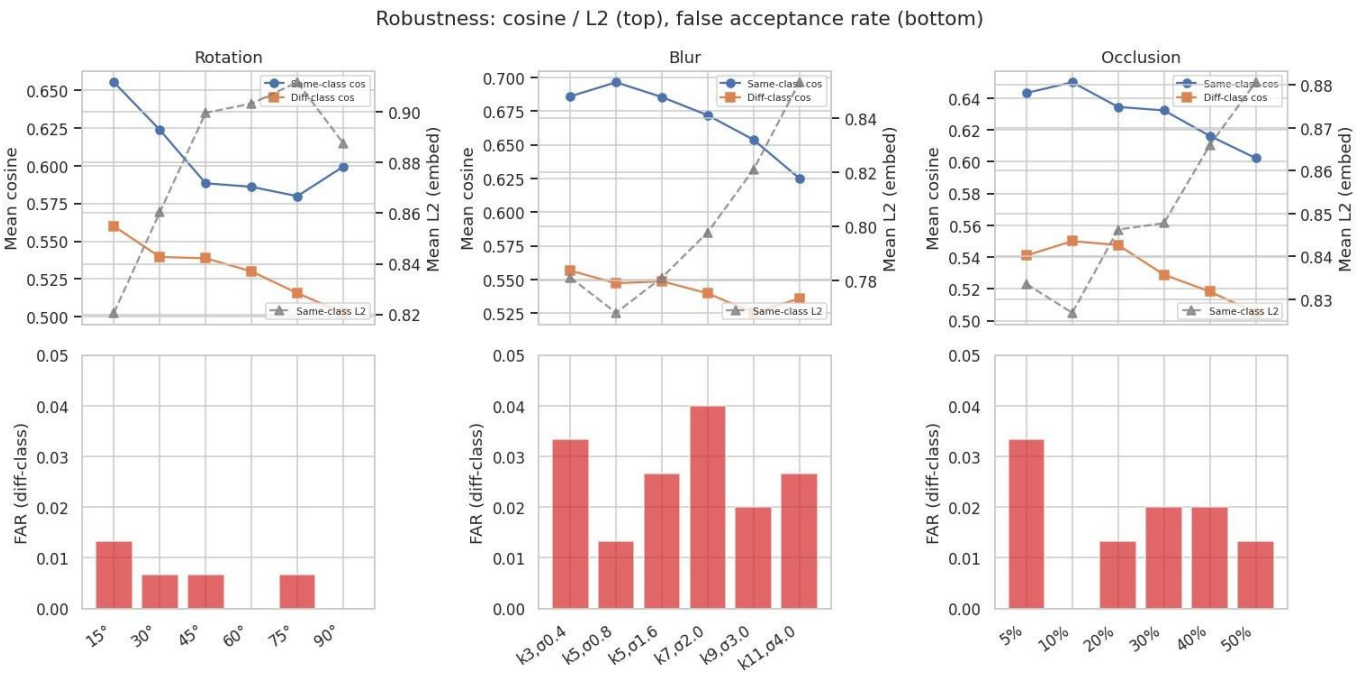


Figure 5. Robustness curves - cosine/L2 metrics (first row) and false acceptance rate (second row) for all three types of perturbations and severities.

Finding 1 - Rotation causes the least damage (AUC: 0.873 to 0.774). Cosine similarity within the same class drops only ~0.05 even at 90° rotation, and FAR stays below 1.5%. This is greater robustness than would be expected of a rotation-invariant model.

> ***Why this happens:*** *ImageNet objects are photographed from many angles and orientations. At pretraining time, the network was implicitly encouraged to learn the same representation of an object at different orientations to minimise cross-entropy loss. This accidental rotation invariance - not architectural symmetry - is why embeddings are robust to rotation. We observe an effect of data-driven invariance acquisition.*

Finding 2 - Blur is the most operationally dangerous (peak FAR 4.0%). Same-class cosine similarity drops from 0.69 to 0.63, but even more concerning is the spike in FAR at moderate blur (k=7, sigma=2.0) rather than at maximum severity - see the non-monotonic curve in Figure 5. The AUC drops from 0.856 to 0.720.

Why this happens: *ResNet-50 encodes obstacles largely using high-frequency texture cues - edges, surface normals, fine textures - that distinguish, say, a pothole (rough asphalt texture) from a road (smooth). These cues are low-pass filtered by Gaussian blur. The non-monotonic FAR at intermediate (not maximum) blur occurs because excessive blur drives all images towards the same featureless grey blob and therefore destroys both in-class and between-class similarity - it is the middle regime where colour and shape are partly obliterated but sufficient remains to spuriously match across classes.*

Finding 3 - Occlusion causes the largest AUC drop (0.875 to 0.700, delta=0.175). At 50% occlusion, half the content of the discriminative patches is removed. Interestingly, the FAR at low occlusion (10-20%) is low, implying that the degradation in the embedding is gradual until a critical point is reached.

Why this happens: *In contrast to blur - which degrades all pixels - occlusion eliminates a region. The embedding is a global average pooling of the features following the final conv layer. With 50% of the patch missing, the pooling is dominated by the background region, which is more similar across classes (asphalt, sky, grass) than the obstacle. The graceful degradation at low occlusion stems from global average pooling distributing mass across all unoccluded regions, so minimal occlusions have minimal effect.*

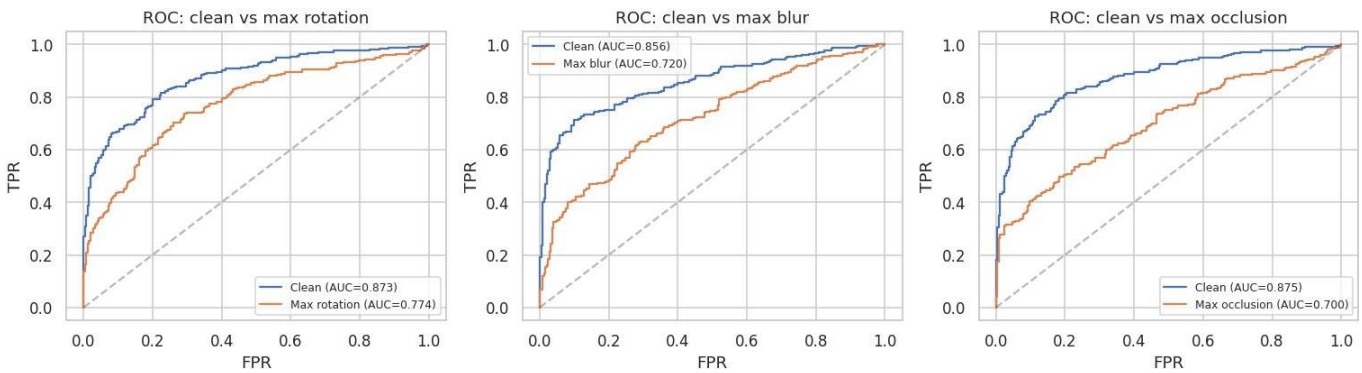


Figure 6. ROC curves: clean vs. max-severity for each perturbation. AUC drops: rotation 0.099, blur 0.136, occlusion 0.175.

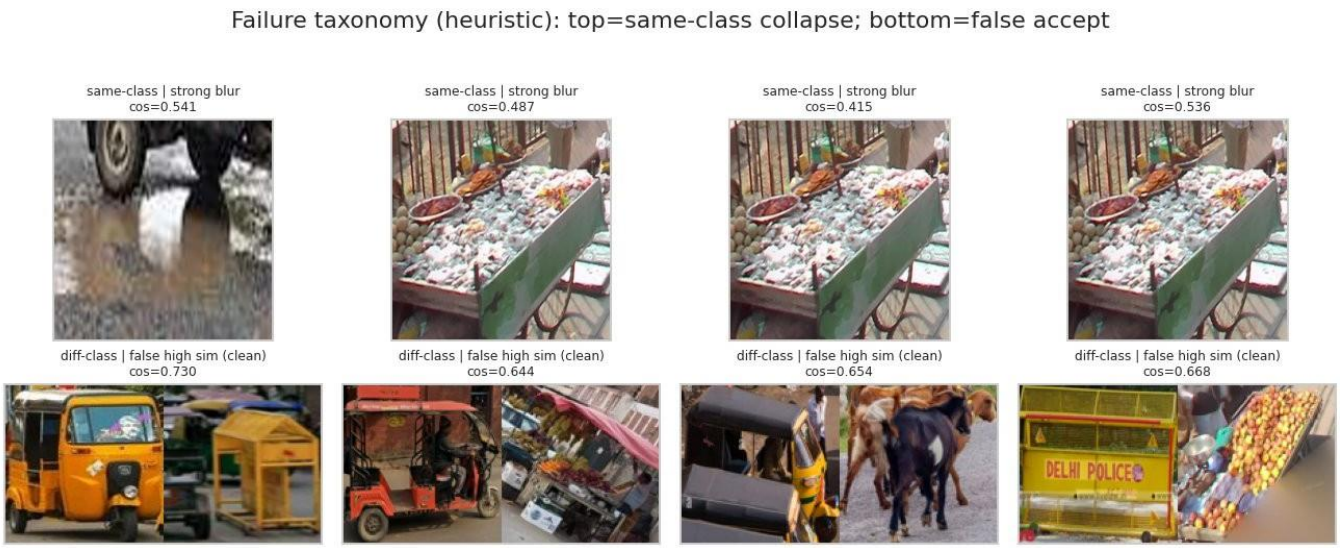


Figure 7. Failure taxonomy. Top: class collapse with heavy blur (cos 0.41-0.54). Bottom: clean false accepts where scene context (road, vegetation) pushes different-class embeddings together.

Finding 4 - Clean false accepts encode scene features. The failure taxonomy (Figure 7, bottom row) shows that even in the absence of any perturbation, some pairs from different classes score higher than the threshold for positive pairs (cos 0.64-0.73). These are visually complex scenes (rickshaw next to thela, animal next to market stall) where the crops share a common background (colour, texture) rather than the obstacle itself.

Why this happens: *The backbone was trained on full ImageNet scenes where the background is informative (a polar bear is likely on snow; a rickshaw is likely on a road). When cropped patches share common background features across classes, the embedding learns common background features instead of obstacle-specific features. This is an example of shortcut learning: the model has learnt to focus on correlations in the training distribution that do not generalise to the similarity task.*

Perturbation	Clean AUC	Max AUC	AUC Drop	Peak FAR	Primary mechanism
Rotation	0.873	0.774	0.099	~1.3%	Mild - implicit ImageNet invariance
Blur	0.856	0.720	0.136	~4.0%	Texture cue destruction (high-freq loss)
Occlusion	0.875	0.700	0.175	~3.3%	Content erasure - background dominance

Table 2. Comparing controlled perturbations - damage severity: Occlusion >= Blur >> Rotation.

Conclusion: Occlusion >= Blur >> Rotation is consistent with the mechanisms: rotation is opposed by implicit pretraining invariance; blur removes texture cues to the embedding; and occlusion removes part of the content completely and displays context dependence. When placed on the roads of India, blur (due to high speed) and partial occlusion (overlapping vehicles) are the challenges, and focused augmentation with these corruptions in fine-tuning would counter them.

Unit III — Transfer Learning - CNN Freezing (Problem 14)

Research question: Which layers should you freeze when fine-tuning?

Experimental Setup

2,542 cropped objects were extracted from YOLO bounding boxes (80/20 train/val). A pretrained ResNet-50 with a replaced 5-class head was fine-tuned for 5 epochs (Adam, learning rate 1e-4, cross-entropy loss) using four strategies. This is a controlled experiment because the model, data, optimizer, learning rate and number of epochs are all the same - only which parameter groups are allowed to receive gradients changes.

Strategy	Frozen Layers	Trainable Layers	Final Acc	Time
Full Train	None	All	98.43%	140.2s
Freeze Early ★	conv1, layer1, layer2	layer3, layer4, fc	98.63%	107.0s
Freeze Late	layer4, fc	conv1-layer3	97.84%	139.7s
Head Only	All conv layers	fc only	96.67%	71.9s

Table 3. Four freezing strategies - controlled experiment on same model, data and training protocol. ★ = best overall.

Results & Mechanistic Interpretation

Finding 1 - Freeze Early is the best strategy (98.63%, -24% time). It is more accurate than Full Train and being substantially faster. This is initially surprising: why does training fewer parameters lead to better accuracy?

> ***Why this happens:*** *ResNet initial layers (conv1, layer1, layer2) learn domain-invariant features: Gabor-like edge detectors, colour blobs, simple texture patterns. These work for ImageNet objects and Indian road obstacles - the underlying statistics of natural images are identical. Fine-tuning these layers on a small dataset risks losing these general features when not enough data is available to re-learn them, leading to noise into the gradient signal. Freezing them eliminates this noise but allows layer3/layer4 - which encode object-specific, higher-level features - to adapt to the new domain. The accuracy gain over Full Train is small but significant, and saves time since gradients are not computed for the frozen layers.*

Finding 2 - Freeze Late is very slow (-0.4%) and less accurate. Training early layers while freezing layer4 and the head is not a good strategy.

> ***Why this happens:*** *Layer4 has the most task-specific features in ResNet. Freezing it means the highest features, which need to be mapped to the obstacle classes, are not tuned to obstacles but are tuned to ImageNet (dogs, cars and furniture) rather than obstacles. The lower layers that are being trained cannot fix this: they are feeding into a frozen high-level representation that is not relevant for the task. The poor time saving comes from backpropagation needing to compute gradients through the frozen layer4 to get to the unfrozen early layers - the parameters in the frozen layer do not get updated but are still part of the backward pass.*

Observation 3 - Head Only begins at 71% (epoch 1) but ends at 96.7%. The slow start and recovery are both mechanistically informative.

Why this happens: *A randomly initialised linear head on top of frozen ImageNet features creates a severe distribution mismatch at epoch 1: the features are tuned to 1000 ImageNet classes, and the random weights make close to random predictions for 5 obstacle classes. As training goes, the head learns which linear combinations of the 2048 frozen features best separate obstacle classes - and because ImageNet features are general and informative, there are good linear separators (96.7% is high for a linear probe). The recovery confirms that ImageNet pretraining is indeed useful for the task; the plateau is an initialisation problem, not a capacity problem.*

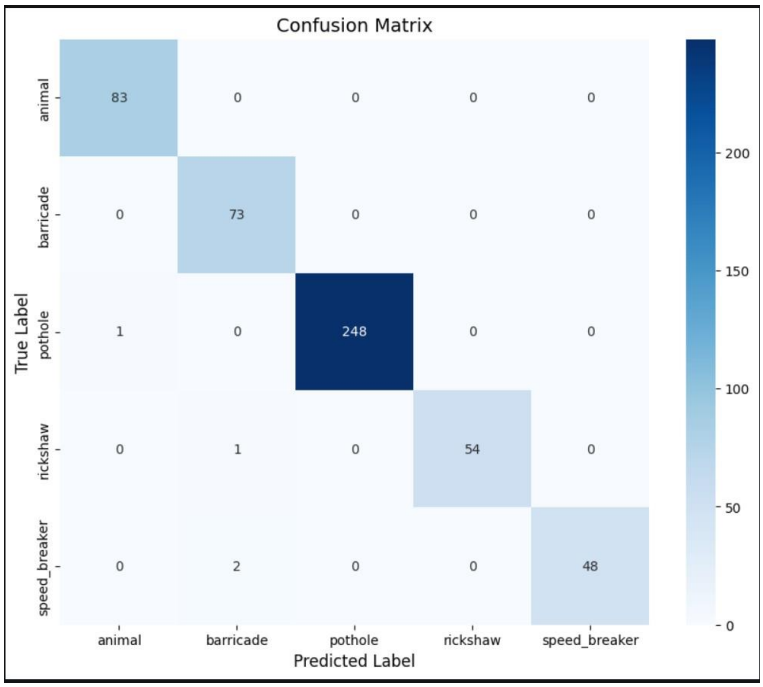


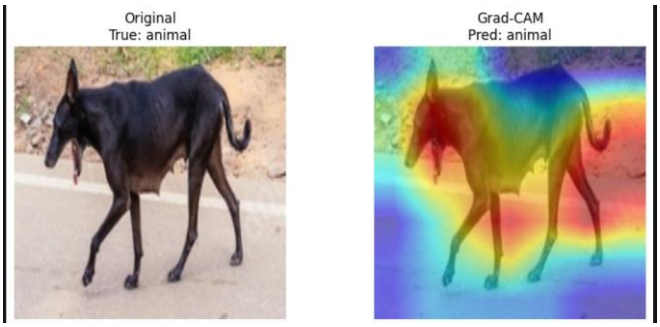
Figure 8. Confusion matrix (Freeze Early model). Just 3 misclassifications out of 510. The 2 speed_breaker to barricade confusions are visually similar: both are horizontal barriers.

Finding 4 - Matrix errors are mechanistically explainable. The two most frequent confusion errors - speed_breaker to barricade and pothole to animal - are not accidental. Barricades share horizontal line structure spanning the road; shadows of dark-coloured animals on the road are similar to dark-coloured potholes.

Why this happens: *Layer4 of ResNet-50 represents shapes and coarse texture combinations, not semantic categories. If two obstacle classes have similar shape statistics at the scale of a 179 px RF, the resulting feature vectors will be near each other in the embedding space - which will lead to systematic confusions along visually motivated class boundaries. This is a property of the representation.*

Grad-CAM - Is the network looking at the right places?

GradCAM on layer4[-1] of the Freeze Early model shows the model is focusing on obstacle structures, and not on background context.



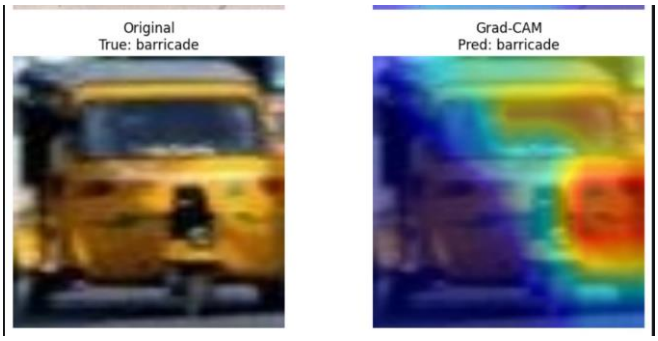
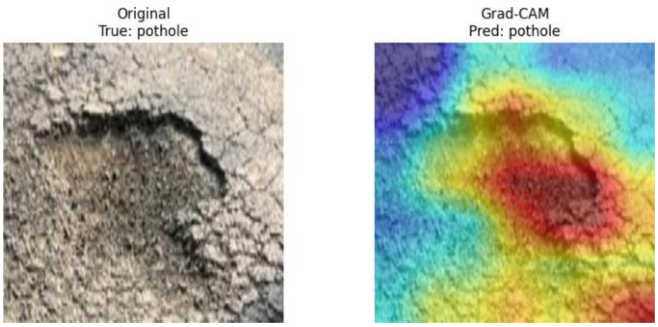


Figure 9. Animal - the body is activated. | Figure 10. Barricade - focus on the structure.



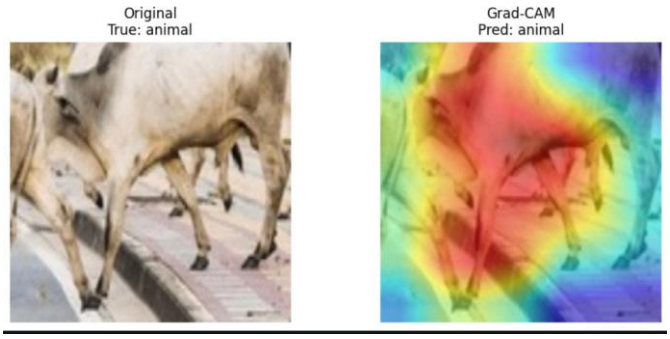


Figure 11. Rickshaw - attention on chassis and wheels. | Figure 12. Pothole - attention on the depression.

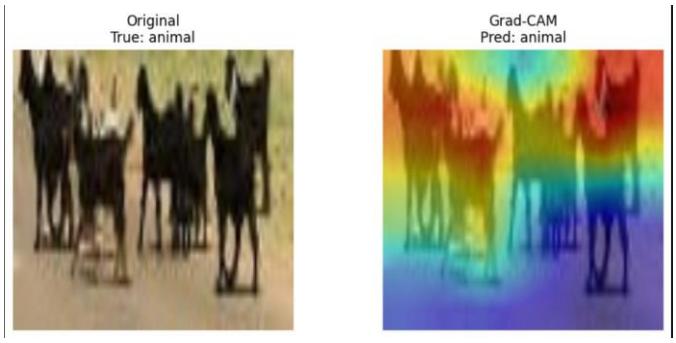


Figure 13. Speed Breaker - attention on the horizontal raised surface.

All GradCAM maps focus on the object body and not the background, showing the model has learned real visual discriminators. This is important: high accuracy on a small dataset could in principle arise from dataset shortcuts (background colour, image framing). GradCAM rules this out.

> ***Why this happens:*** *With fine-tuning of layer3/layer4/fc (Freeze Early), the gradient signal is focused on which high-level spatial features (already partially learnt in the frozen lower layers) discriminate the five classes. Lower layers being frozen, the model cannot "cheat" by going back through the background statistics, and has to learn to use higher-level object-specific features. This is why Freeze Early generates more class-discriminative attention than a fully-trained model with more freedom to exploit shortcuts.*

Conclusion

Freeze Early is the best approach for small datasets near domain boundaries: it preserves low-level general features from being wiped out, trains high-level layers to the new domain, and cuts training time by 24% - a significant time-saving in the development cycle. Full Train is worth the additional cost only if a validation set shows overfitting.

Cross-Cutting Analysis

Our three experiments tell a coherent mechanistic tale about the application of ImageNet-pretrained CNNs to Indian road obstacles. Problem 1 found early pooling irrevocably destroys high-frequency information, RF depth determines global sensitivity, and that skip connections distribute representations. Problem 7 then showed that the structure in the embedding space is invariant to orientation (because ImageNet training induces rotation invariance) but is sensitive to blur and occlusion because these attack the exact features (texture, object-part content) that Problem 1 highlighted as depth sensitive. Problem 14 closed the loop: the layered representation we probed in Problem 1 readily explains the success of Freeze Early - early layers contain the domain-general features that can be transferred, but later layers contain the domain-specific features that need re-tuning. The GradCAM findings in Problem 14 also support the receptive field predictions in Problem 1: it focuses on object shape for big obstacles and surface texture for potholes, as predicted by the RF-predicted sensitivity profile.